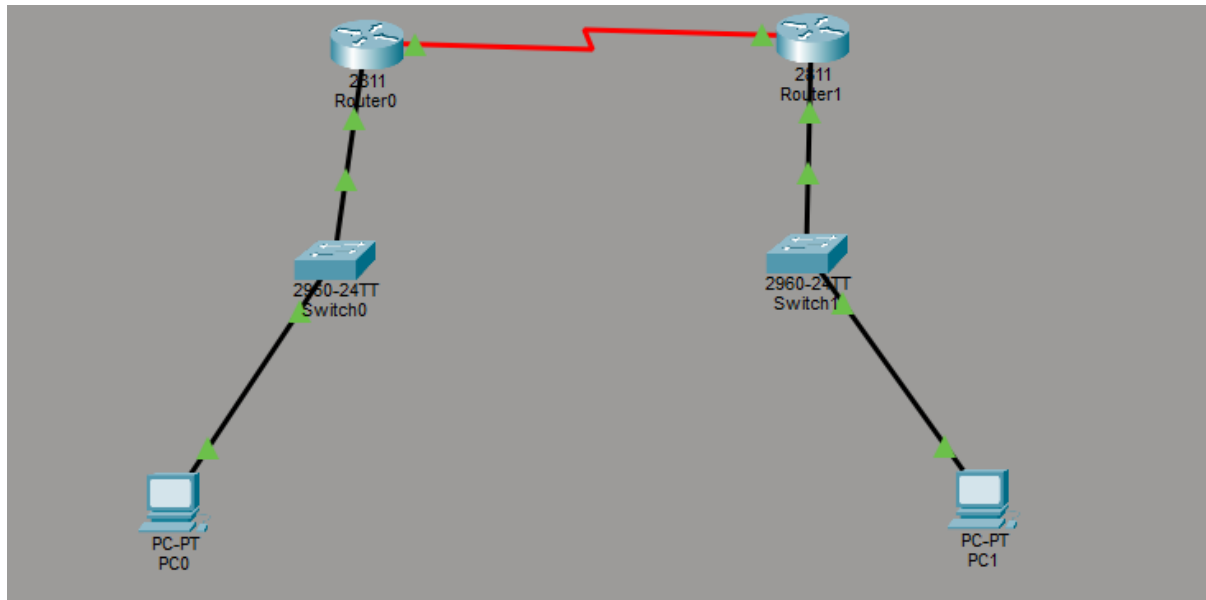


Assignment 5

Ques1)

Topology-



Configuration Commands on Router 1-

```
Router>en
Router#confi t
^
% Invalid input detected at '^' marker.

Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface fastEthernet 0/1
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
ex
Router(config)#interface serial 0/2/1
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#clock rate 64000
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial0/2/1, changed state to down
Router(config-if)#ex
Router(config)#
%LINK-5-CHANGED: Interface Serial0/2/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/2/1, changed state to up
Router(config)#ex
```

```
Router#show ip interface brief
Interface          IP-Address      OK? Method Status      Protocol
FastEthernet0/0    unassigned      YES unset   administratively down down
FastEthernet0/1    192.168.1.1     YES manual   up            up
Serial0/2/0        unassigned      YES unset   administratively down down
Serial0/2/1        10.0.0.1        YES manual   up            up
Vlan1              unassigned      YES unset   administratively down down
```

```
Router#show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.0.0.0/30 is directly connected, Serial0/2/1
L    10.0.0.1/32 is directly connected, Serial0/2/1
192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/24 is directly connected, FastEthernet0/1
L    192.168.1.1/32 is directly connected, FastEthernet0/1
```

```
Router>en
```

```
Router#config t
```

```
Enter configuration commands, one per line. End with CNTL/Z.
```

```
Router(config)#ip route 172.16.0.0 255.255.0.0 10.0.0.2
```

```
Router(config)#ex
```

```
Router#
```

```
%SYS-5-CONFIG_I: Configured from console by console
```

```
Router#show ip route
```

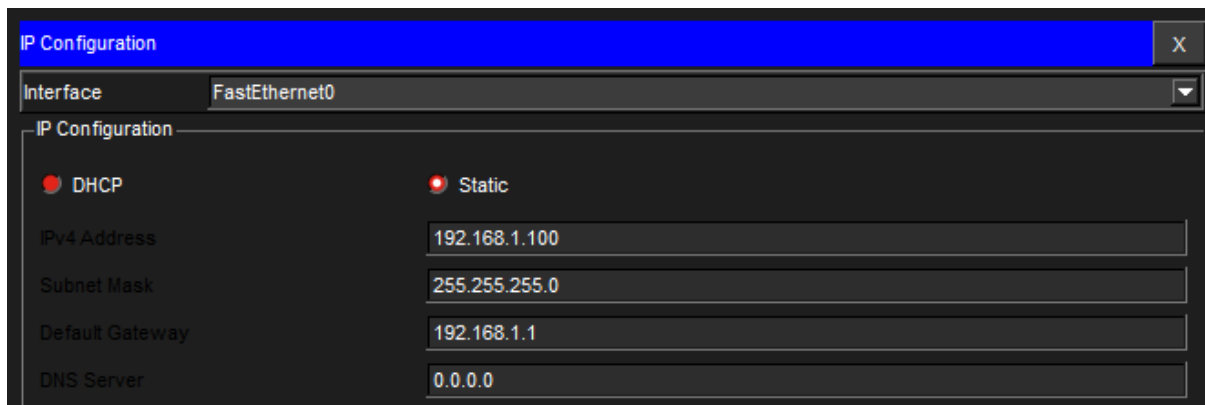
```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.0.0.0/30 is directly connected, Serial0/2/1
L    10.0.0.1/32 is directly connected, Serial0/2/1
S    172.16.0.0/16 [1/0] via 10.0.0.2
192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/24 is directly connected, FastEthernet0/1
L    192.168.1.1/32 is directly connected, FastEthernet0/1
```

```
Router#
```

Configuration Commands on PC1-



IP Configuration	
Interface	FastEthernet0
IP Configuration	
☒ DHCP	☐ Static
IPv4 Address	192.168.1.100
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DNS Server	0.0.0.0

Configuration Commands on Router 2-

```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface fastEthernet 0/1
Router(config-if)#ip address 172.16.0.1 255.255.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
ex
Router(config)#interface serial 0/0/1
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/1, changed state to up
ex
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/1, changed state to up
Router(config)#ex
```

```
Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 unassigned      YES unset   administratively down down
FastEthernet0/1 172.16.0.1      YES manual   up          up
Serial0/0/0      unassigned      YES unset   administratively down down
Serial0/0/1      10.0.0.2        YES manual   up          up
Vlan1           unassigned      YES unset   administratively down down
```

```
Router#show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/0/1
L       10.0.0.2/32 is directly connected, Serial0/0/1
      172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.0.0/16 is directly connected, FastEthernet0/1
L       172.16.0.1/32 is directly connected, FastEthernet0/1
```

```
Router>en
```

```
Router#config t
```

```
Enter configuration commands, one per line. End with CNTL/Z.
```

```
Router(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1
```

```
Router(config)#ex
```

```
Router#
```

```
%SYS-5-CONFIG_I: Configured from console by console
```

```
Router#show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/0/1
L       10.0.0.2/32 is directly connected, Serial0/0/1
      172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.0.0/16 is directly connected, FastEthernet0/1
L       172.16.0.1/32 is directly connected, FastEthernet0/1
S       192.168.1.0/24 [1/0] via 10.0.0.1
```

Configuration Commands on PC2-

IP Configuration

Interface: FastEthernet0

IP Configuration

☒ DHCP ☒ Static

IPv4 Address: 172.16.0.100

Subnet Mask: 255.255.0.0

Default Gateway: 172.16.0.1

DNS Server: 0.0.0.0

Ping from PC1-

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.0.100

Pinging 172.16.0.100 with 32 bytes of data:

Request timed out.
Reply from 172.16.0.100: bytes=32 time=38ms TTL=126
Reply from 172.16.0.100: bytes=32 time=24ms TTL=126
Reply from 172.16.0.100: bytes=32 time=30ms TTL=126

Ping statistics for 172.16.0.100:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 24ms, Maximum = 38ms, Average = 30ms
```

Ping from PC2-

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.100

Pinging 192.168.1.100 with 32 bytes of data:

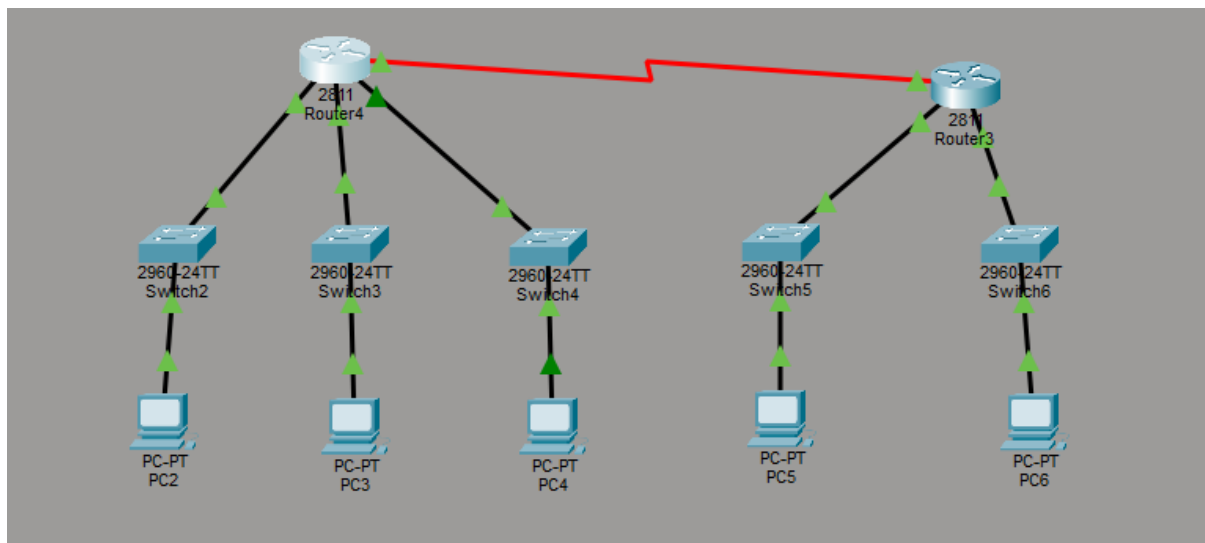
Reply from 192.168.1.100: bytes=32 time=11ms TTL=126
Reply from 192.168.1.100: bytes=32 time=28ms TTL=126
Reply from 192.168.1.100: bytes=32 time=37ms TTL=126
Reply from 192.168.1.100: bytes=32 time=12ms TTL=126

Ping statistics for 192.168.1.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 11ms, Maximum = 37ms, Average = 22ms

C:\>
```

Ques2)

Topology-



Configuration on Router 4-

```
Router#show ip interface brief
Interface          IP-Address      OK? Method Status      Protocol
FastEthernet0/0    198.168.1.1     YES manual up          up
FastEthernet0/1    198.168.1.65    YES manual up          up
Serial10/0/0       unassigned      YES manual down        down
Serial10/0/1       10.0.0.1        YES manual up          up
FastEthernet1/0    198.168.1.129   YES manual up          up
FastEthernet1/1    unassigned      YES unset  administratively down down
Vlan1              unassigned      YES unset  administratively down down
Router#

Router#
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.0.0.0/30 is directly connected, Serial10/0/1
L    10.0.0.1/32 is directly connected, Serial10/0/1
198.168.1.0/24 is variably subnetted, 6 subnets, 2 masks
C    198.168.1.0/26 is directly connected, FastEthernet0/0
L    198.168.1.1/32 is directly connected, FastEthernet0/0
C    198.168.1.64/26 is directly connected, FastEthernet0/1
L    198.168.1.65/32 is directly connected, FastEthernet0/1
C    198.168.1.128/26 is directly connected, FastEthernet1/0
L    198.168.1.129/32 is directly connected, FastEthernet1/0
220.168.2.0/25 is subnetted, 2 subnets
S    220.168.2.0/25 [1/0] via 10.0.0.2
--More-- |
```

PC2 configuration –

Interface	FastEthernet0
IP Configuration	
☒ DHCP	☒ Static
IPv4 Address	198.168.1.10
Subnet Mask	255.255.255.0
Default Gateway	198.168.1.1
DNS Server	0.0.0.0

PC3 configuration –

Interface	FastEthernet0
IP Configuration	
☒ DHCP	☒ Static
IPv4 Address	198.168.1.70
Subnet Mask	255.255.255.0
Default Gateway	198.168.1.65
DNS Server	0.0.0.0

PC4 configuration –

IP Configuration	
☒ DHCP	☒ Static
IPv4 Address	198.168.1.130
Subnet Mask	255.255.255.0
Default Gateway	198.168.1.129
DNS Server	0.0.0.0

Ping from PC2 (subnet 1)-

```

C:\>ping 220.168.2.10

Pinging 220.168.2.10 with 32 bytes of data:

Reply from 220.168.2.10: bytes=32 time=10ms TTL=126
Reply from 220.168.2.10: bytes=32 time=12ms TTL=126
Reply from 220.168.2.10: bytes=32 time=14ms TTL=126
Reply from 220.168.2.10: bytes=32 time=24ms TTL=126

Ping statistics for 220.168.2.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 24ms, Average = 15ms

C:\>

```

Configuration on Router 3-

```

Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface fa0/0
Router(config-if)#ip address 220.168.2.1 255.255.255.128
Router(config-if)#ex
Router(config)#interface fa0/1
Router(config-if)#ip address 220.168.2.129 255.255.255.128
Router(config-if)#ex
Router(config)#interface serial 0/2/0
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#ex
Router(config)#ip route 198.168.1.0 255.255.255.192 10.0.0.1
Router(config)#ip route 198.168.1.64 255.255.255.192 10.0.0.1
Router(config)#ip route 198.168.1.128 255.255.255.192 10.0.0.1
Router(config)#

```

```

Router>
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface fa0/0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#ex
Router(config)#interface fa0/1
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

```



```

Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface serial 0/2/0
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shut

Router(config-if)#
%LINK-S-CHANGED: Interface Serial0/2/0, changed state to up

Router(config-if)#
%LINEPROTO-S-UPDOWN: Line protocol on Interface Serial0/2/0, changed state to up

```

```

Router>en
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

```

Gateway of last resort is not set

```

      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/2/0
L       10.0.0.2/32 is directly connected, Serial0/2/0
      198.168.1.0/26 is subnetted, 3 subnets
S       198.168.1.0/26 [1/0] via 10.0.0.1
S       198.168.1.64/26 [1/0] via 10.0.0.1
S       198.168.1.128/26 [1/0] via 10.0.0.1
      220.168.2.0/24 is variably subnetted, 4 subnets, 2 masks
C       220.168.2.0/25 is directly connected, FastEthernet0/0
L       220.168.2.1/32 is directly connected, FastEthernet0/0
C       220.168.2.128/25 is directly connected, FastEthernet0/1
L       220.168.2.129/32 is directly connected, FastEthernet0/1
--More--

```

PC5 configuration –

IP Configuration	
☒ DHCP	☒ Static
IPv4 Address	220.168.2.10
Subnet Mask	255.255.255.0
Default Gateway	220.168.2.1
DNS Server	0.0.0.0

PC6 configuration –

IP Configuration	
☒ DHCP	☒ Static
IPv4 Address	220.168.2.130
Subnet Mask	255.255.255.0
Default Gateway	220.168.2.129
DNS Server	0.0.0.0

Pinging from PC5-

```
C:\>ping 198.168.1.10

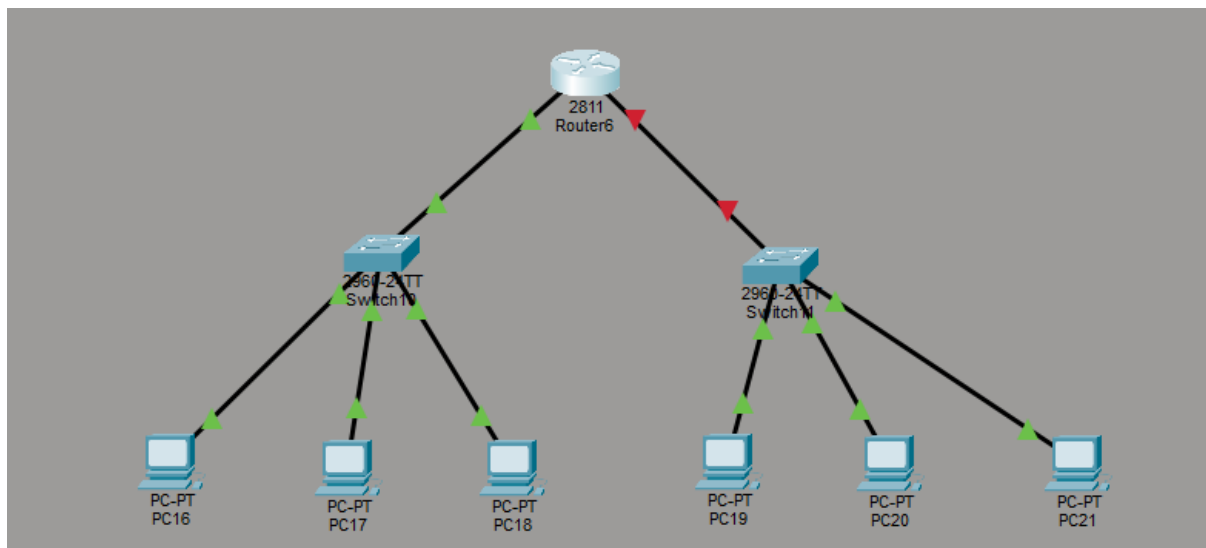
Pinging 198.168.1.10 with 32 bytes of data:

Reply from 198.168.1.10: bytes=32 time=11ms TTL=126
Reply from 198.168.1.10: bytes=32 time=14ms TTL=126
Reply from 198.168.1.10: bytes=32 time=10ms TTL=126
Reply from 198.168.1.10: bytes=32 time=12ms TTL=126

Ping statistics for 198.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 14ms, Average = 11ms
```

Ques3)

Topology-



Configuration on Router6-

```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface fa0/0
Router(config-if)#ip address 192.168.3.1 255.255.255.248
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#ex
Router(config)#interface fa0/1
Router(config-if)#ip address 224.152.1.1 255.255.255.248
Not a valid host address - 224.152.1.1
Router(config-if)#
```

"224.x.x.x belongs to multicast address range, so it cannot be used for host addressing.

"The given network 224.152.1.0 belongs to Class D (multicast), which is not valid for host communication.